

TECHNICAL BULLETIN

DIXIE CHEMICAL
10601 Bay Area Blvd.
Pasadena, TX 77507
Tel: (281) 474-3271
Fax: (281) 291-3384
E-Mail: info@dixiechemical.com

Nonenylsuccinic Anhydride (NSA)

CAS Reg. Nr. 92077-08-2

Typical Properties

Property	Values	Test Method
Physical Appearance	Clear Yellow Liquid	D-104 (Visual)
Specific gravity @ 25 °C, g/cc	1.029	D-102 (ASTM D 4052)
Gardner Color	6 Max	D-150G
Neut. Number, mg KOH/g	487-507	D-150I

About

Nonenyl Succinic Anhydride (NSA) is produced from the reaction of maleic anhydride with propylene trimer (C9 olefin). The resulting product is a low viscosity liquid anhydride that can be used in a number of applications.

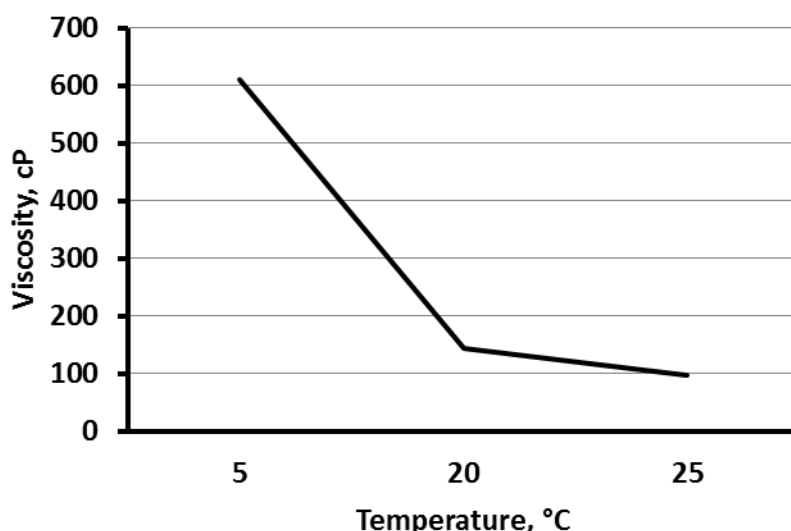
Uses

NSA can be used as an epoxy curative alone or in blends with other anhydrides such as DDSA. Epoxy blends with NSA have low working viscosity for ease of use. The low viscosity and moderate density makes NSA an attractive choice for curing epoxy-based syntactic foams. NSA finds application in fuel and lubrication additives, surfactants and intermediates for other chemistries.

TECHNICAL BULLETIN

NSA

Viscosity as a Function of Temperature



NSA has the advantage of low viscosity over the range of 5 – 25°C. At low temperatures, similar products such as DDSA are much higher in viscosity. This could translate into easier handling at low temperatures without having to warm tanks to pump out material.

Example Epoxy Formulations

Standard bisphenol-A epoxy (Epon 828 from Momentive) was formulated with NSA or DDSA curative at the appropriate ratio, as shown in the table below. Viscosity of the uncured resin was tested at 25°C. Samples were then cured for 30 minutes at 100°C and then 30 minutes at 160°C and glass transition temperature (T_g) was determined.

Characteristic	NSA	DDSA
Blend Composition	100 parts EPON 828 114 parts NSA 1.5 parts BDMA	100 parts EPON 828 135 parts DDSA 1.5 parts BDMA
Viscosity of uncured epoxy mixture @ 25°C, cP	610	1195
Cured T_g	72.5°C	48.5°C

TECHNICAL BULLETIN

NSA

In these tests, NSA cured epoxy displayed lower uncured viscosity and higher cured T_g than DDSA cured epoxy.

For more information on the use of anhydrides like NSA as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical Company.

Handling & Storage

NSA will react with water to form nonenylsuccinic acid. This is normally undesirable, so NSA should be stored in such a way that it is carefully protected from moisture contamination.

Safety Precautions

Read and understand the relevant Material Safety Data Sheets (MSDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the MSDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the MSDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

Nonenylsuccinic anhydride is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.